CURRICULUM OVERVIEW

Advanced Data Science

Build 15+ Real AI & Data Science Projects

A complete, project-driven journey from your first line of Python to arcade games, data dashboards and machine learning models you can deploy. Students don't just learn about AI and data. They write the code, build the projects and prove it with tests.



15+ REAL AI & DATA SCIENCE PROJECTS
Portfolio-ready work

6 TESTS TO PROVE IT
5 course tests + 1 final test on the whole curriculum

5	40	46	12–18
Courses	Lessons	Sessions in total	Age range

WHO IT IS FOR
Beginner friendly — starts from scratch

BEFORE YOU START
No coding experience needed

LEARNING PROGRESSION

01 Python for Young Builders
From block coding to real Python. Build 3 apps and a mini game.

02 Python Arcade Games
Pygame, sprites and collisions. Ship 3 playable arcade games.

03 Python for Data
Pandas, cleaning and statistics on real datasets.

04 Data Visualisation
Charts, Plotly and Dash dashboards that tell a story.

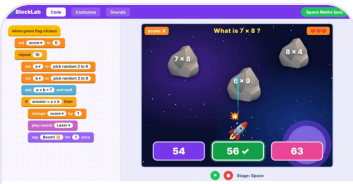
05 Machine Learning Projects
Train, tune and deploy models that predict and recommend.

Learn by building, not by watching
Every lesson pairs a real idea with a real, hands-on build. No shallow demos, no unfinished prototypes.

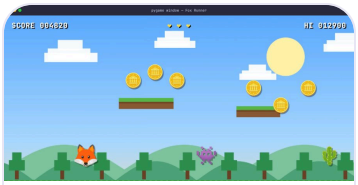
A portfolio, not just a certificate
Students finish with real projects they can show, explain and keep building on.

Tested, so progress is real
Every course ends with a test, and a final test closes the program, so progress is proven, not assumed.

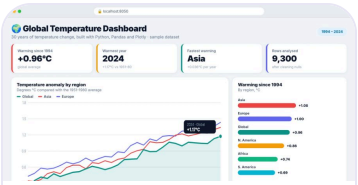
SAMPLE HANDS-ON PROJECTS STUDENTS BUILD



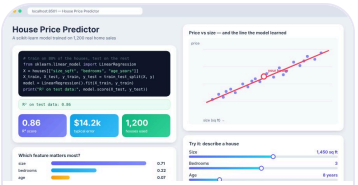
01 Space Maths Quiz
Python · Variables · Decisions



02 Python Arcade Game
Python · Pygame



04 Climate Change Dashboard
Python · Plotly · Dash



05 House Price Predictor
Python · scikit-learn

COURSE 01 · LESSONS 1–8

01 Python for Young Builders

Go from block coding to real Python by building your first apps and games

Python 3 | Variables & Input | Loops & Lists | Mini Games

3 REAL PROJECTS

1 COURSE TEST

OVERVIEW

The foundation of everything. Your child moves from blocks to real Python: variables, decisions, loops and lists, and finishes with a mini game of their own. No fluff, just real code.

PORTFOLIO PROJECTS

- Personal Quiz App
- Simple Calculator
- Word Guessing Game

WHY IT MATTERS

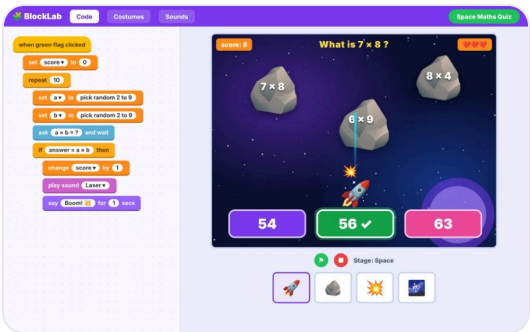
Every program, from a quiz app to an AI model, is built from these same building blocks. Students learn to think in code, break problems down and ship small projects that actually run.

8	3	1
LESSONS	PROJECTS	COURSE TEST

LESSON BREAKDOWN

1. From Blocks to Words — Hello, Python!
2. Variables and User Input
3. Printing and String Formatting
4. If/Else Decisions
5. While Loops and For Loops
6. Lists and Simple Data
7. Building a Mini Quiz App
8. Final Project — Our Own Mini Game!

8 LESSONS — TEST UNLOCKS



SAMPLE HANDS-ON PROJECT

Space Maths Quiz

Python · Variables · Decisions

The kind of first app students build: a quiz that asks questions, checks answers and keeps score — all in their own Python.

COURSE TEST

Python for Young Builders — Course Test

Checks what you built and learned in this course. Taken after lesson 8, before the next course begins.

30 multiple-choice | 10 true / false | 40 marks

COURSE 02 · LESSONS 9–16

02 Python Arcade Games

Build playable arcade games with Pygame and share them with the world

Python · Pygame · Sprites & Sound · Game Loop

3 REAL PROJECTS

1 COURSE TEST

OVERVIEW

Your child builds real arcade games: a space shooter, a maze runner and a brick breaker. They learn how game loops, collisions, enemies and scoring really work, then package and share their games.

PORTFOLIO PROJECTS

- Space Shooter
- Maze Runner with Timer
- Brick Breaker Clone

WHY IT MATTERS

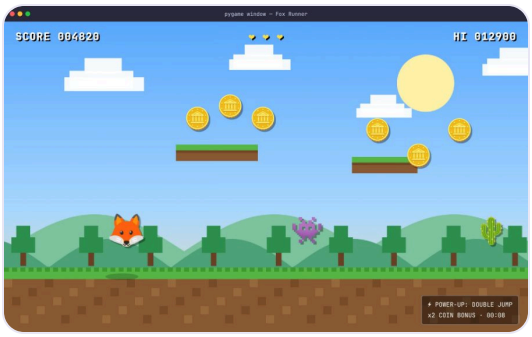
Games are the most motivating way to master programming. Students practise logic, maths and problem-solving every time they tweak a level, and finish with games friends and family can play.

8	3	1
LESSONS	PROJECTS	COURSE TEST

LESSON BREAKDOWN

1. Python Basics for Game Development
2. Pygame Setup and the Game Loop
3. Sprites, Images, and Movement
4. Collision Detection
5. Enemies, AI Behavior, and Scoring
6. Sound Effects and Music
7. Game States — Menu, Play, Game Over
8. Packaging and Sharing Your Game

8 LESSONS — TEST UNLOCKS



SAMPLE HANDS-ON PROJECT

Python Arcade Game

Python · Pygame

A side-scrolling arcade game with enemies, coins and a high-score table — built from the game loop up.

COURSE TEST

Python Arcade Games — Course Test

Checks what you built and learned in this course. Taken after lesson 8, before the next course begins.

30 multiple-choice · 10 true / false · 40 marks

COURSE 03 · LESSONS 17–24

03 Python for Data

Turn raw data into answers with Pandas and real-world datasets

Python · Pandas · Jupyter Notebooks · CSV & Excel

3 REAL PROJECTS

1 COURSE TEST

OVERVIEW

Your child learns how data scientists work: loading real data, cleaning it, grouping it and finding patterns. They analyse sports, school and city datasets, the same skills used in business and research.

PORTFOLIO PROJECTS

- Sports Performance Analysis
- School Grade Analysis
- City Population Trends

WHY IT MATTERS

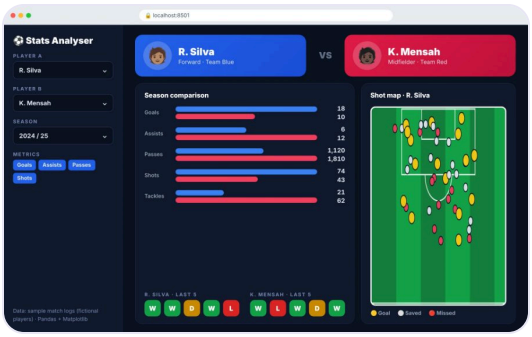
Data drives decisions in every industry. Students learn the full analysis workflow, from messy files to clear insights, which is the first step of every data science and AI project.

8	3	1
LESSONS	PROJECTS	COURSE TEST

LESSON BREAKDOWN

1. Why Data Science Matters (Real Examples)
2. Python Refresher and Jupyter Setup
3. Loading Data from CSV and Excel
4. Pandas — Selecting, Filtering, Sorting
5. Cleaning Data — Nulls, Duplicates, Types
6. Aggregations — Group By, Pivot Tables
7. Statistical Summaries and Correlation
8. Full Analysis Project with a Real Dataset

8 LESSONS — TEST UNLOCKS



SAMPLE HANDS-ON PROJECT

Sports Stats Analyser

Python · Pandas

A player-comparison tool built from a real season of sports data: loaded, cleaned, grouped and compared with Pandas.

COURSE TEST

Python for Data — Course Test

Checks what you built and learned in this course. Taken after lesson 8, before the next course begins.

30 multiple-choice · 10 true / false · 40 marks

COURSE 04 · LESSONS 25–32

04 Data Visualisation

Create charts and interactive dashboards that tell a clear story

Matplotlib · Seaborn · Plotly · Dash

3 REAL PROJECTS

1 COURSE TEST

OVERVIEW

Your child learns to make data visible: from classic charts to interactive dashboards. They build a climate dashboard, a market trends chart and a sports comparison tool that anyone can explore.

PORTFOLIO PROJECTS

- Climate Change Dashboard
- Market Trends Chart
- Sports Comparison Visualiser

WHY IT MATTERS

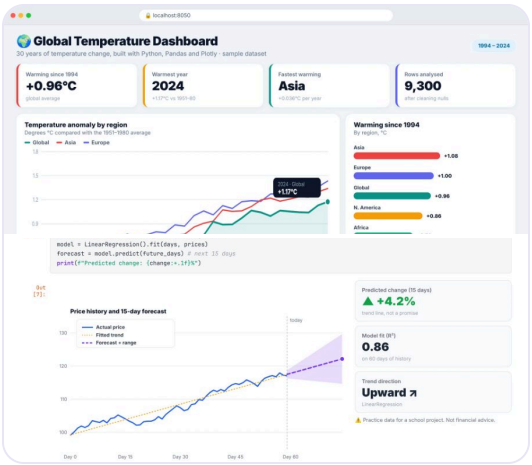
A good chart can change minds. Students learn which chart fits which data, how colour and labels shape the message, and how to present findings to people who are not technical.

8	3	1
LESSONS	PROJECTS	COURSE TEST

LESSON BREAKDOWN

1. The Science of Good Data Visualisation
2. Matplotlib: Line, Bar, Scatter Plots
3. Seaborn: Statistical Visualisations
4. Choosing the Right Chart for Your Data
5. Color, Labels, and Storytelling
6. Interactive Charts with Plotly
7. Building a Dash Dashboard
8. Your Own Interactive Dashboard

8 LESSONS — TEST UNLOCKS



SAMPLE HANDS-ON PROJECTS

Climate Change Dashboard · Market Trends Forecast

Python · Plotly · Dash | Python · Matplotlib

Charts that tell a story: an interactive climate dashboard and a market trend chart with a forecast line.

COURSE TEST

Data Visualisation — Course Test

Checks what you built and learned in this course. Taken after lesson 8, before the next course begins.

30 multiple-choice · 10 true / false · 40 marks

COURSE 05 · LESSONS 33–40

05 Machine Learning Projects

Train, test and deploy models that predict, classify and recommend

Python · scikit-learn · Pandas · Model APIs

3 REAL PROJECTS

1 COURSE TEST

OVERVIEW

Your child learns how machine learning really works, with no maths jargon. They train models that predict house prices, spot customers about to leave and recommend movies, then deploy one as an API.

PORTFOLIO PROJECTS

- House Price Predictor
- Customer Churn Predictor
- Movie Recommendation Engine

WHY IT MATTERS

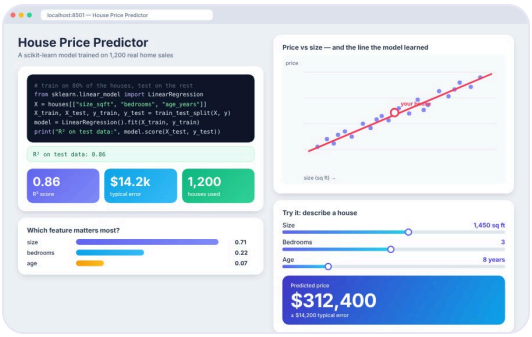
Every ML system, from spam filters to recommendation engines, follows the same workflow: prepare data, train a model, test it honestly and explain the result. This course puts it all together.

8	3	1
LESSONS	PROJECTS	COURSE TEST

LESSON BREAKDOWN

1. How Machine Learning Works (No Maths Jargon)
2. Types of ML — Regression vs Classification
3. Train/Test Splits and Model Evaluation
4. Linear Regression From Scratch
5. Decision Trees and Random Forests
6. Feature Engineering and Selection
7. Model Tuning and Improving Accuracy
8. Deploying a Prediction Model as an API

8 LESSONS — TEST UNLOCKS



SAMPLE HANDS-ON PROJECT

House Price Predictor

Python · scikit-learn

A model trained on real housing data that predicts a price from size, rooms and age — with its accuracy shown honestly.

COURSE TEST

Machine Learning Projects — Course Test

Checks what you built and learned in this course. Taken after lesson 8, before the next course begins.

30 multiple-choice · 10 true / false · 40 marks

HOW STUDENTS ARE TESTED · 6 TESTS

Level-Up Checkpoints

Five course tests and one final test turn “I watched” into “I can”. Each course ends with a checkpoint on what that course taught, and the final test pulls the whole curriculum together.



THE CHECKPOINT PATH — EVERY COURSE ENDS WITH A TEST, THEN ONE FINAL

Course 1 · 8 lessons → COURSE 1 → Course 2 · 8 lessons → COURSE 2 → Course 3 · 8 lessons → COURSE 3 → Course 4 · 8 lessons → COURSE 4 → Course 5 · 8 lessons → COURSE 5 → FINAL

COURSE TEST · every 8 lessons

Show what you built and learned

After every 8 lessons a course test checks the ideas and skills of that course, so each next course starts on solid ground.

- **30** multiple-choice questions — choose one of four answers
- **10** true / false questions
- **40** marks in total — 1 mark for every question

FINAL TEST · the finish line

One final test on the whole curriculum

One final test across Python, games, data, visualisation and machine learning: questions plus a small practical project.

- **25** multiple-choice questions
- **5** true / false questions
- **20** marks for one small practical project (about 15 minutes) that your teacher marks

A TASTE OF A TEST

SAMPLE · MULTIPLE-CHOICE

Why do we keep some of the data aside as a test set?

- A To make the model train faster
- B To check the model on examples it has never seen ✓ CORRECT
- C To delete the rows we do not like
- D To make the chart colourful

SAMPLE · TRUE / FALSE

A chart’s colours and labels can change the message it sends.

True ✓ | False

Examples in the style of a real test — written for this document, not taken from one.

AFTER EVERY TEST

Saved as you go
Answers save automatically while a student works, so nothing is lost if the connection drops.

Checked by the teacher
The teacher reviews the report and shares the result with the student.

Fair for everyone
Questions and answer options appear in a different order for every student.

The Complete Journey

Across 5 courses, 40 lessons and 6 tests, students progress from their first Python program to building, visualising and deploying real AI and data science projects, finishing with 15+ working projects for their portfolio.

5 COURSES · 40 LESSONS

01 Python for Young Builders
Personal Quiz App · Simple Calculator · Word Guessing Game
8 lessons · + course test

02 Python Arcade Games
Space Shooter · Maze Runner with Timer · Brick Breaker Clone
8 lessons · + course test

03 Python for Data
Sports Performance Analysis · School Grade Analysis · City Population Trends
8 lessons · + course test

04 Data Visualisation
Climate Change Dashboard · Market Trends Chart · Sports Comparison Visualiser
8 lessons · + course test

05 Machine Learning Projects
House Price Predictor · Customer Churn Predictor · Movie Recommendation Engine
8 lessons · + course test

HOW STUDENTS ARE TESTED · 6 TESTS

COURSE TEST 1 — Python for Young Builders
COURSE TEST 2 — Python Arcade Games
COURSE TEST 3 — Python for Data
COURSE TEST 4 — Data Visualisation
COURSE TEST 5 — Machine Learning Projects

FINAL TEST
One final test on the whole curriculum
One final test across Python, games, data, visualisation and machine learning: questions plus a small practical project.

BY THE END, STUDENTS CAN

- Write real Python programs and build playable games from scratch.
- Clean, analyse and visualise real data with Pandas, Matplotlib and Plotly.
- Train, test and deploy machine learning models as a working API.

Ready to start building?
All lessons and projects are taught live by an expert Lumexa instructor.
Lumexa Advanced Data Science · Ages 12–18